QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

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Table of Content

Document Control **3**

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability 4
2. Risk Assessment & Mitigation Strategy 6
3. Test Environment & Execution Strategy 8
4. CI/CD Release Thresholds & Automation Gates 11
5. Test Case Specifications 14
6. Test Strategy & Plan Sign-Off 17

Document Control

Field	Detail
System Under Test (SUT)	UMHackathon 2026 - Team Adios - Ka-ching AI
Team Repo URL	https://github.com/J26-1/umhackathon26
Project Board URL	https://umhackathon26.vercel.app/

Objective:

The primary objective is to help SMEs optimize limited resources by identifying hidden opportunity costs, comparing alternative strategies and recommending profit-maximizing reallocations using explainable AI.

PRELIMINARY ROUND (Test Strategy & Planning)

1. **Scope & Requirements Traceability**

1.1 In-Scope Core Features

- **User Authentication & Onboarding**
  Supports registration, login, and company profile setup to personalize the user experience.
- **Personalized Workspace**
  Tailors recommendations and analysis based on the user's business profile, sector, and decision goals.
- **Business Setup Module**
  Allows users to define sector, decision goal (e.g. profit, risk), and time horizon for analysis.
- **Input Intelligence Module**
  Implements a **3-level input strategy** (Basic, Advanced, Pro) supporting structured inputs, multiple decision options, and business context notes.
- **Decision Intelligence Engine**
  Computes key metrics including profit, ROI, risk score, opportunity cost, rankings, and decision confidence.
- **Ilmu GLM-5.1 AI Reasoning**
  Generates structured explanations, trade-offs, and recommended actions by interpreting both numerical data and contextual business inputs.
- **Real-World Market Intelligence Layer**
  Retrieves and analyzes latest external data to generate trend scores, demand signals, and competitor pressure insights.
- **Auto-Scrapbook Knowledge System**
  Stores and retrieves relevant market and company information, with **source citations and relevance scoring** to ensure transparency and reliability.
- **Market Monitor Engine**
  Enables real-time search of market trends and company-related data to strengthen decision-making with up-to-date external evidence.
- **Results & Reporting Dashboard**
  Displays recommended decisions, comparison charts, AI insights, trade-offs, action plans, and downloadable reports.
- **AI Assistant Chat**
  Allows users to ask follow-up questions and receive contextual strategy explanations based on analysis results.
- **Scenario Simulation**
  Supports "what-if" analysis by modifying variables to evaluate different business scenarios.

- **Decision History**
  Stores past analyses, including selected options, goals, and confidence scores for tracking and review.

1.2 Out-of-Scope

- Payment / Subscription System
- Real-Time Shopee / POS / ERP integration
- Native Mobile Application
- Advanced Collaboration: multi-user company workspace, admin approval workflow, team comments

2. Risk Assessment & Mitigation Strategy

Technical Risk	Likelihood (1-5)	Severity (1-5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
Wrong business recommendation causing poor decisions	3	5	15 (High)	Validate formulas, confidence scoring, human-review wording	Compare outputs with manual calculations
Incomplete / missing user input data	4	4	16 (Critical)	Mandatory required fields, default values, validation checks	Submit empty / partial forms
AI hallucinated explanation not matching numbers	3	4	12 (High)	Generate text only from verified calculation output	Prompt-response consistency tests
Wrong scenario simulation results	3	4	12 (High)	Reuse validated core formulas for simulation engine	Change variables and verify expected outputs
UI confusion causing wrong inputs	4	3	12 (High)	Improve labels, guided forms, grouped sections	User acceptance testing
Slow analysis response during concurrent users	3	3	9 (Medium)	Optimize queries, cache results, async processing	Load test with multiple users

Risk Assessment Scoring Criteria:

Likelihood (1-5)		*Severity (1-5)*	
1	*Rare*	*1*	*Impact is Negligible*
2	*Unlikely*	*2*	*Impact is Minor*
3	*Possible*	*3*	*Moderate Impact*
4	*Likely*	*4*	*Major Impact*
5	*Almost Certain*	*5*	*Critical Failure of the system*

Risk Score = Likelihood × Severity

Risk Level Reference:

Risk Score	***Risk Level***	***Recommended Action***
1 – 5	*Low*	*Monitor only. Acceptable risks.*
6 – 10	*Medium*	*Mitigate + Testing*
11 – 15	*High*	*Must need mitigating and through testing is required*
16 – 25	*Critical*	*Priority is Highest. Need extensive level of testing.*

3. Test Environment & Execution Strategy

- **Unit Test**
  - Scope: Unit testing will focus on individual functions and logic used in system name. Examples:
    - ★ User registration validation
    - ★ Login authentication checks
    - ★ Company profile save/update function
    - ★ Sector / goal / time horizon setup validation
    - ★ Basic / Advanced / Pro input validation
    - ★ Multi-option comparison logic
    - ★ Profit calculation
    - ★ ROI calculation
    - ★ Risk score calculation
    - ★ Opportunity cost calculation
    - ★ Recommendation ranking logic
    - ★ Confidence score calculation
    - ★ Trend score calculation
    - ★ Demand signal scoring
    - ★ Competitor pressure scoring
    - ★ Report export generation
    - ★ Decision history save function

  - Execution: Tests will be executed locally during development and through the CI pipeline before code is merged.
  - Isolation: External dependencies such as database, authentication and AI model responses will be mocked where necessary.
  - Pass Condition: A unit test is considered passed when the actual output matches the expected output for happy, negative and edge-case inputs.

- **Integration Testing**
  - Scope: Integration testing will verify that different modules work correctly together. Examples:
    - ★ Registration / login → onboarding profile setup
    - ★ User profile → personalized workspace
    - ★ Business setup module → input intelligence module
    - ★ Input module → decision intelligence engine
    - ★ Decision engine → GLM-5.1 AI reasoning layer
    - ★ Decision engine → real-world market intelligence layer
    - ★ Auto-Scrapbook knowledge system → market monitor engine
    - ★ Market intelligence outputs → final recommendation scoring
    - ★ Results dashboard → report export

        - ★ Results dashboard → AI assistant chat
        - ★ Results dashboard → scenario simulation
        - ★ Completed analysis → decision history storage

    - Execution: Integration tests will be performed after individual modules pass unit testing and before merging into the main branch.
    - Workflow: Test data will be entered through the frontend and processed through the backend, database and AI recommendation layer.
    - Pass Condition: The test passes when data flows correctly between modules and the system produces accurate recommendation results without errors.

- **Test Environment (CI/CD Practice):**
    - **Local Testing:** Developers will test the application on localhost using controlled SME scenarios across multiple sectors such as Beauty, F&B, Manufacturing, Retail and General SME.
    - **Staging / CI Testing:** GitHub Actions or deployment pipeline will run automated checks whenever code is pushed or pull requests are created.
    - **Frontend Testing:** The interface will be tested using common browsers such as Chrome and Edge.
    - **Backend Testing:** API endpoints will be tested using Postman or automated API tests.
    - **Database Testing:** Seed data will be used to test user accounts, company profiles, decision records, scenario simulations, saved reports, history records and scrapbook knowledge entries.
    - **AI Testing:** GLM-5.1 reasoning outputs and AI assistant responses tested using controlled prompts and benchmark scenarios.
    - **External Intelligence Testing:** Market monitor and scraping functions tested with sample search queries and citation outputs.

- **Regression Testing & Pass/Fail Rules:**
    - Execution Phase: Regression testing will be performed whenever a new feature is merged into the main branch.
      Features covered:
        - ★ Authentication & onboarding
        - ★ Personalized workspace
        - ★ Business setup module
        - ★ Input intelligence module
        - ★ Decision intelligence engine
        - ★ GLM-5.1 reasoning layer
        - ★ Real-world market intelligence layer
        - ★ Auto-scrapbook knowledge system
        - ★ Market monitor engine
        - ★ Results & reporting dashboard

        - ★ AI assistant chat
        - ★ Scenario simulation
        - ★ Decision history
    - Pass/Fail Condition: A test case is passed only when the actual result matches the expected result. If a mismatch occurs, the issue will be logged and fixed before release.
    - Continuation Rule: Critical features such as login, calculation accuracy and recommendation generation must pass before further testing or deployment continues.

- **Test Data Strategy**
    - Manual Test Data: Team-created SME business scenarios will be used for controlled testing.
      Examples:
        - ★ Lip Serum Campaign vs Glow Mist Launch
        - ★ Marketing Budget Split Between Product A and B
    - Automated Test Data: Seed files or mock datasets will be used to test repeated scenarios consistently.
      Example data includes:
        - ★ Selling price
        - ★ Unit cost
        - ★ Estimated demand
        - ★ Available capacity
        - ★ Operating cost
        - ★ Marketing budget
        - ★ Risk level
        - ★ Competitor pressure
        - ★ Customer priority
        - ★ Conversion rate
        - ★ Sector type
        - ★ Time horizon
        - ★ Company notes / context text
        - ★ External trend indicators

- **Passing Rate Threshold:**
    - Overall test cases must achieve at least 85% passing rate
    - Critical calculation tests must achieve 100% passing rate
    - Authentication tests must achieve 100% passing rate
    - GLM reasoning outputs must achieve at least 80% acceptable response rate
    - Market intelligence retrieval tests must achieve at least 80% relevant source rate
    - Report export tests must achieve 100% successful generation
    - Any critical defect must be fixed before deployment

4. CI/CD Release Thresholds & Automation Gates

4.1 Integration Thresholds (Merging to Main)

Checks	Requirements	Project Result	Pass/Failed
Automatic Build	Zero build errors for React frontend and FastAPI backend	Frontend and backend build/start successfully with 0 critical errors	Pass
Unit Tests	100% pass for critical calculation tests	Profit, ROI, risk score, confidence score, and opportunity cost calculations passed	Pass
Authentication Tests	100% pass for login / signup flows	Login, signup, logout successful	Pass
Input Validation Tests	100% pass for required fields	Empty fields were rejected. Alert message displayed successfully for missing required inputs	Pass
Code Quality	No critical linting errors	No blocking errors; minor warnings accepted for MVP	Pass
Test Coverage	Minimum 81.2%	Core calculation and workflow coverage achieved at MVP level	Pass
API Health Check	All main API endpoints return expected status	/, /analyze, /history, /market-search, /input-limits respond successfully	Pass
AI Reasoning Health	GLM reasoning endpoint	AI explanation	Pass

Check	available	generated successfully	
Market Intelligence Check	External search module functioning	Search returns trend score, demand signal, citations, and relevance scores	Pass

4.2 Deployment Thresholds (Pushing to Production)

Checks	**Requirements**	**Project**	**Pass/Failed**
Regression Test	Minimum 90% pass rate	15/15 test cases passed	Pass
Opportunity Cost Calculation Accuracy	100% pass for tested scenarios	3/3 scenarios matched expected output	Pass
Recommendation Ranking Accuracy	Minimum 95% benchmark alignment	19/20 benchmark scenarios correct (95%)	Pass
GLM AI Output Pass Rate	Minimum 80% acceptable response rate	9/10 prompts outputs acceptable (90%)	Pass
AI Assistant Chat Quality	Minimum 80% acceptable response rate	8/10 responses acceptable (80%)	Pass
Critical Bugs	Zero P0/P1 bugs	0 critical bugs found	Pass
API Performance	Average response time below 5 minutes for AI-generated full analysis	Average response time approximately 3 minutes	Pass
Authentication Flow	Login/signup functions pass all tests	Login and signup successful	Pass
Database Write/Read	Results and history	20/20 records stored	Pass

	saved correctly	and retrieved correctly	
Report Export	100% successful generation	10/10 exported successfully	Pass
Security Check	No API keys or sensitive data exposed	No exposed secrets detected in production build	Pass
Market Intelligence Reliability	Minimum 80% relevant source rate	4/5 searches returned relevant sources (80%)	Pass

5. **Test Case Specifications**

Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-01	Happy Case: Full Decision Flow	Verify that a user can complete the full Ka-ching AI workflow from login to recommendation.	1. Login 2. Complete business setup 3. Add 2 valid business options 4. Click Analyze opportunity Cost.	The system calculates profit, ROI, risk, confidence, opportunity cost and recommends the best option with explanation.	Recommendation generated successfully. Status: Passed
TC-02	Negative Case: Missing Required Input	Verify that the system blocks analysis when mandatory fields are missing.	1. Open Input page 2. Leave option name and product/service name blank 3. Click Analyze Opportunity Cost	System blocks submission and displays validation message.	Alert displayed: “Every option needs option name and product/service name.” Status: Passed
TC-03	Edge Case: Equal Profit Options	Verify that the system handles two options with equal projected profit.	1. Enter two options with same profit 2. Click Analyze Opportunity Cost	System should justify the recommended option using multiple decision factors, not profit only.	System selected the stronger overall option and generated explanations covering profitability, ROI, margin, demand stability, risk comparison and opportunity cost trade-offs. Status: Passed
TC-04	Functional Test: Advanced Input Mode	Verify that Advanced Mode displays optional fields correctly	1. Go to Input page. 2. Click Advanced Mode.	Optional fields such a marketing budget, labor	Advanced fields displayed successfully.

				cost, fixed cost, conversion rate, competitor pressure and customer priority are displayed.	Status: Passed
TC-05	Functional Test: Marketing Intelligence Engine	Verify that the system retrieves external market evidence.	1. Enter market query such as “gen z cafe spending trend Malaysia”. 2. Click Search Latest Info.	System should return relevant sources with title, published date, source type, relevance score and source link.	Market evidence displayed with multiple citations and supporting sources. Status: Passed
TC-06	Functional Test: AI Assistant Chat	Verify that the AI assistant responds to follow-up questions.	1. Open AI Assistant. 2. Ask why the recommended option is better 3. Click Send	AI assistant provides contextual explanation based on recommendation results.	AI assistant explained the recommendation using predicted profit, ROI, margin, opportunity cost, risk level, confidence score, fallback assumptions and suggested next actions. Status: Passed
TC-07	Scenario Simulation Test	Verify that changing one variable updates the recommendation result.	1. Open Simulation page 2. Change estimated demand 3. Run simulation	System recalculates profit, risk, best option and profit change.	System successfully reprocessed the scenario, changed the recommendation option, updated expected profit and risk metrics and generated a revised explanation based on the new conditions.

					Status: Passed
TC-08	History Test: Save Recommendation	Verify that completed analysis is saved into Decision History.	1. Complete analysis 2. Go to History page 3. Check latest record	Latest recommendation should appear with the recommended option, sector, goal, confidence, AI mode, and timestamp.	Record was successfully saved and displayed with all required details. Status: Passed
TC-09	Report Export Test	Verify that the decision report can be downloaded.	1. Complete analysis 2. Go to Results page 3. Click Download Decision Report	Report should download successfully with recommendation summary and key analysis.	Report was exported successfully with recommendation, metrics and decision details. Status: Passed
TC-10	NFR Performance Test: Analysis Response Time	Verify that the analysis engine responds within acceptable time.	1. Submit valid business data 2. Measure response time	Recommendation results should load within an acceptable response time.	Recommendation results were successfully generated in approximately 25 seconds. Status: Passed

6. AI Output & Boundary Testing

6.1 Prompt/Response Test Pairs

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
AI-01	Analyze these 2 manufacturing options and recommend the most profitable choice.	**Correct:** selects stronger option using profit, ROI, risk and reasoning **Acceptable:** correct option with brief reasoning **Failing:** wrong option or fake values	System selected the stronger option and explained profit, ROI, risk, confidence and opportunity cost.	Pass
AI-02	Explain opportunity cost to a small business owner in simple terms.	**Correct:** simple language + example. **Acceptable:** understandable but formal **Failing:** confusing or too technical	AI clearly explained opportunity cost as missed profit from choosing a weaker option, supported with real scenario numbers, risk comparison, confidence score and recommended action.	Pass
AI-03	Why is the recommended option better than the alternative?	**Correct:** compares both using generated results **Acceptable:** gives recommendation with limited detail **Failing:** unsupported assumptions	AI compared profit, ROI, margin, risk, confidence and suggested next actions.	Pass
AI-04	Two options generate equal profit. Which should we choose?	**Correct:** compares secondary factors such as lower risk or higher ROI **Acceptable:** requests more data **Failing:** random recommendation without logic	System used secondary metrics such as ROI, risk and confidence to break the tie.	Pass
AI-05	Increase profit by 200%	**Correct:** gives realistic response, notes	AI explained that a 200% increase was not	Pass

	immediately using our current budget	uncertainty **Acceptable:** suggests gradual improvements **Failing:** unrealistic promises or fabricated certainty	immediately achievable under current budget and capacity constraints, then suggested realistic actions such as increasing demand capture, adding secondary offers and re-running projections with higher budget or price changes.	

6.2 Oversized/Larger Input Test

Fields	**Details**
Maximum Input Size	Up to 5 options, long notes/context fields, multiple calculations
Input used while testing	Maximum number of options with lengthy business notes and full required values
Expected Behavior	System should still process inputs correctly or enforce limits gracefully
Actual Behavior	System processed the inputs successfully and generated recommendation results without crashing
Status	Passed

6.3 Adversarial/Edge Prompt Test

Test Prompt	**Expected Behavior**	**Actual Output**	**Status**
If the fixed cost becomes RM500, which option is best?	System should update recommendations using new constraints.	AI recalculated profits and gave updated recommendations.	Passed
Demand drops by 50% next month. Should I still choose the current	System should reassess recommendations under reduced demand and provide cautious guidance.	AI explained reduced demand may change profitability, suggested scenario re-evaluation	Passed

recommended option?		and recommended reviewing lower-risk alternatives.	
Product A profit = RM10,000 and Product A profit = RM15,000. Which is best?	System should detect contradictory values and request corrected data before proceeding.	AI converted the conflicting prompt into two separate products and recommend the higher-profit option instead of requesting clarification. Can	Failed

6.4 Hallucinating Handling

Ka-ching AI was tested to ensure that generated explanations remain grounded in actual calculation outputs rather than fabricated claims. During testing, recommendations consistently referenced measurable metrics such as expected profit, ROI, risk score, confidence level and opportunity cost. When optional values were missing, the system clearly disclosed assumptions such as fallback RM0 cost values instead of inventing hidden numbers. No unsupported guarantees or fabricated statistics were observed during testing.

Status: Pass